

JAYA SRAVAN KALA

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EDUCATION

George Mason University <i>Master of Science in Computer Science</i> Relevant Courses: Machine Learning, Data Mining, Analysis of Algorithms, Natural Language Processing	August 2022 – May 2024 Fairfax, VA
Institute Of Aeronautical Engineering <i>Bachelor of Technology in Information Technology</i>	August 2017 – May 2021 Hyderabad, India

WORK EXPERIENCE

Sam's Club <i>Computer Vision Engineer</i> - Developed production YOLO-based OBB, segmentation, and detection models on retail inventory datasets, achieving 97% mAP across critical product classes; tracked all experiments with MLflow for full reproducibility. - Stabilized segmentation outputs via VSB-guided bin splitting, NMS, and mask IoU filtering, reducing duplicate/overlapping detections by ~40% across 100+ retail camera streams. - Designed automated image ingestion pipeline integrating Verint API with quality filters (blur, fisheye detection), increasing dataset diversity by ~35% and reducing model retraining frequency by 2x. - Implemented pseudo-labeling workflow publishing pre-annotated datasets to TrueSpace via Azure Service Bus, cutting manual labeling time by ~50% and accelerating annotation cycles from weeks to days. - Built inventory change detection system combining IoU-based spatial alignment, DOLG embedding similarity (cosine distance), and multimodal confirmation for robust product-level change validation. - Developing similarity-based UPC assignment using rolling DOLG template embeddings from historical labeled crops to replace coordinate-dependent mapping.	February 2025 – Present Remote
UCode Technologies LLC <i>Software Trainee</i> - Built ETL pipelines to extract and clean raw data from SQL databases and AWS S3, transforming it into structured, model-ready datasets for predictive analytics. - Automated manual data preparation workflows, enabling the data science team to run supervised ML models on business performance data. - Trained and evaluated classification models including Random Forest, tuning precision/recall thresholds to align with business requirements. - Performed data quality checks and handled missing/inconsistent values to ensure clean feature inputs for model training.	September 2024 – January 2025 Ashburn, VA
Cognizant <i>Program Analyst Trainee</i> - Executed and monitored scheduled production ETL workflows across MySQL and Snowflake environments, ensuring data availability and SLA adherence. - Performed data cleaning and SQL-based transformation to prepare high-quality datasets for downstream analytics dashboards. - Supported Snowflake schema design and warehouse data structuring for analytics and reporting use cases.	February 2021 – July 2022 Hyderabad, India

PROJECTS

Multimodal Fake News Detection <i>CLIP, ViLT, PyTorch, HuggingFace</i> - Built multimodal ML system to detect inconsistencies between article text and images using CLIP and ViLT embeddings. - Implemented cross-modal similarity comparison using cosine distance to identify misleading text-image pairs. - Developed attention-based interpretability analysis to inspect and visualize model predictions.
Formula 1 Data Analytics Pipeline <i>Azure Data Factory, PySpark, Delta Lake</i> - Designed end-to-end data pipeline using Azure Data Factory, PySpark, and Delta Lake to process multi-season race, driver, and constructor datasets. - Built batch and incremental ingestion workflows with schema enforcement and optimized partitioning to support time-series analytics.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, JavaScript
AI & Computer Vision: PyTorch, YOLOv11 (OBB, Segmentation, Detection), Ultralytics, OpenCV, HuggingFace Transformers, DOLG Embeddings, TorchScript, Scikit-learn
MLOps & Deployment: MLflow (Model Registry & Versioning), ONNX Runtime, Model Quantization, CI/CD Pipelines
Cloud & Infrastructure: Azure (Service Bus, Cosmos DB), AWS S3, Google Cloud Vision, Linux/SSH
Data & Engineering: Pandas, NumPy, PySpark, Delta Lake, REST APIs, Unit Testing

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate (DP-203)
- AWS Certified Cloud Practitioner